

Sebastián Andrés Barbieri Lemp

Associate Professor



personal information

www.sbarbieri.cl
sbarbieri@uchile.cl
Born August 1st 1989
Chilean and Italian

Research Interests

I am interested in dynamical systems and ergodic theory. Particularly in topics linking symbolic dynamics with group theory and computability.

Research identifiers

 0000-0001-9567-2085

 MR Author ID=1184309

 barbieri.sebastian

 barbieri_s_1

 Google scholar

SHORT RESUMÉ

2026–	Associate professor DEPARTMENT OF MATHEMATICS · Universidad de Chile
2024–2026	Associate professor DMCC · Universidad de Santiago de Chile
2020–2023	Assistant professor DMCC · Universidad de Santiago de Chile
2019–2020	Postdoctoral researcher LABRI · Université de Bordeaux
2017–2019	Postdoctoral fellow DEPARTMENT OF MATHEMATICS · University of British Columbia
2014–2017	PhD candidate LIP · École normale supérieure de Lyon Thesis: Shift spaces on groups: computability and dynamics.
2008–2014	Mathematical Engineering DIM · Universidad de Chile

RESEARCH GRANTS

2024–2027	Fondecyt regular 1240085 PI · Symbolic dynamics on groups
2021–2023	Fondecyt iniciación 11200037 PI · Computability and symbolic dynamics of group actions
2019–2022	ANR-18-CE40-000 COLLABORATOR · Computer orbits for Discrete Dynamical Systems
2017–2021	ANR-16-CE40-0005 COLLABORATOR · Computational and Combinatorial aspects of Symbolic Dynamics on Groups

PUBLICATIONS IN JOURNALS

- [1] S. Barbieri and N. Carrasco-Vargas. "Medvedev degrees of subshifts on groups". *Theoretical Computer Science* (to appear).
- [2] S. Barbieri, K. Blot, M. Sablik, and V. Salo. "A geometric obstruction to self-simulation for groups". *Discrete and Continuous Dynamical Systems* 59.0 (2027), pp. 492–518.
- [3] S. Barbieri and N. Bitar. "A general framework for quasi-isometries in symbolic dynamics beyond groups". *Journal of Modern Dynamics* 22.0 (2026), pp. 417–467.
- [4] S. Barbieri, F. García-Ramos, and S. Taati. "Cellular automata, percolation and dynamical dichotomies". *Transactions of the American Mathematical Society* 379.9 (2026), pp. 6237–6265.
- [5] S. Barbieri and L. Poirier. "A remark on inverse limits of effective subshifts". *Bulletin of the London Mathematical Society* 58.4 (2026), e70277.
- [6] S. Barbieri, M. Sablik, and V. Salo. "Self-simulable groups". *Transactions of the American Mathematical Society* 379.4 (2026), pp. 2255–2306.
- [7] S. Barbieri, N. Carrasco-Vargas, and C. Rojas. "Effective dynamical systems beyond dimension zero and factors of SFTs". *Ergodic Theory and Dynamical Systems* 45.5 (2025), pp. 1329–1369.
- [8] S. Barbieri, J. Kari, and V. Salo. "The group of reversible turing machines: subgroups, generators, and computability". *Forum of Mathematics. Sigma* 13 (2025), Paper No. e176.
- [9] S. Barbieri and S. Labbé. "Indistinguishable asymptotic pairs and multidimensional Sturmian configurations". *Ergodic Theory and Dynamical Systems* 45.2 (2025), pp. 337–395.
- [10] S. Barbieri, M. Sablik, and V. Salo. "Soficity of free extensions of effective subshifts". *Discrete and Continuous Dynamical Systems* 45.4 (2025), pp. 1117–1149.
- [11] S. Barbieri, R. Bissacot, G. Dalle Vedove, and P. Thieullen. "Zero-temperature chaos in bidimensional models with finite-range potentials". *Advances in Mathematics* 457 (2024), pp. 109906, 1–51.
- [12] S. Barbieri. "Aperiodic subshifts of finite type on groups which are not finitely generated". *Proceedings of the American Mathematical Society* (2023).

- [13] S. Barbieri and T. Meyerovitch. "The Lanford-Ruelle theorem for actions of sofic groups". *Transactions of the American Mathematical Society* 376.2 (2023), pp. 1299–1342.
- [14] S. Barbieri, F. García-Ramos, and H. Li. "Markovian properties of continuous group actions: algebraic actions, entropy and the homoclinic group". *Advances in Mathematics* 397 (2022), pp. 108196, 1–52.
- [15] S. Barbieri. "On the entropies of subshifts of finite type on countable amenable groups". *Groups, Geometry, and Dynamics* 15.2 (2021), pp. 607–638.
- [16] S. Barbieri and F. García-Ramos. "A hierarchy of topological systems with completely positive entropy". *Journal d'Analyse Mathématique* 143.2 (2021), pp. 639–680.
- [17] S. Barbieri, R. Gómez, B. Marcus, T. Meyerovitch, and S. Taati. "Gibbsian representations of continuous specifications: the theorems of Kozlov and Sullivan revisited". *Communications in Mathematical Physics* 382.2 (2021), pp. 1111–1164.
- [18] S. Barbieri, S. Labbé, and Š. Starosta. "A characterization of Sturmian sequences by indistinguishable asymptotic pairs". *European Journal of Combinatorics* 95.103318 (2021), pp. 1–22.
- [19] S. Barbieri, R. Gómez, B. Marcus, and S. Taati. "Equivalence of relative Gibbs and relative equilibrium measures for actions of countable amenable groups". *Nonlinearity* 33.5 (2020), pp. 2409–2454.
- [20] N. Aubrun, S. Barbieri, and S. Thomassé. "Realization of aperiodic subshifts and uniform densities in groups". *Groups, Geometry, and Dynamics* 13.1 (2019), pp. 107–129.
- [21] S. Barbieri. "A geometric simulation theorem on direct products of finitely generated groups". *Discrete Analysis* (2019), Paper No. 9, 25.
- [22] S. Barbieri and M. Sablik. "A generalization of the simulation theorem for semidirect products". *Ergodic Theory and Dynamical Systems* 39.12 (2019), pp. 3185–3206.
- [23] N. Aubrun, S. Barbieri, and M. Sablik. "A notion of effectiveness for subshifts on finitely generated groups". *Theoretical Computer Science* 661 (2017), pp. 35–55.

CONFERENCE PROCEEDINGS

- [24] N. Aubrun, S. Barbieri, and E. Moutot. "The Domino Problem is Undecidable on Surface Groups". *44th International Symposium on Mathematical Foundations of Computer Science (MFCS 2019)*. Vol. 138. 2019, 46:1–46:14.
- [25] S. Barbieri, J. Kari, and V. Salo. "The Group of Reversible Turing Machines". *Cellular Automata and Discrete Complex Systems, Proceedings*. 2016, pp. 49–62.
- [26] S. Barbieri and M. Sablik. "The Domino Problem for Self-similar Structures". *Computability in Europe: Pursuit of the Universal, Proceedings*. 2016, pp. 205–214.

BOOK CHAPTERS

- [27] N. Aubrun, S. Barbieri, and E. Jeandel. "About the Domino Problem for Subshifts on Groups". en. *Sequences, Groups, and Number Theory*. Ed. by V. Berthé and M. Rigo. Trends in Mathematics. Cham: Springer International Publishing, 2018, pp. 331–389.

PREPRINTS

- [28] S. Barbieri, N. Carrasco-Vargas, and P. Toussaint. "Medvedev degrees of SFTs on co-compact Fuchsian groups". *arXiv:2609.04353* (2026).
- [29] S. Barbieri, N. Carrasco-Vargas, and P. Rivera-Burgos. "The automorphism group of a strongly irreducible subshift on a group". *arXiv:2501.14463* (2025).

ADMINISTRATIVE DUTIES

2026–	Organizer of colloquium "las palmeras" · U. DE CHILE
2024–2026	Master committee member · DMCC – USACH
2021–2026	Outreach committee member · FACULTY OF SCIENCE – USACH
2021–2025	Subdirector of outreach · DMCC – USACH
2021–2024	Continuous studies committee member · USACH
2021–2023	Organizer of the DMCC colloquium · DMCC – USACH
2015–2017	Organizer of the MC2 seminar · ENS DE LYON

COURSES TAUGHT

2026-II	Probability theory · MATHEMATICS – U. DE CHILE
2026-I	Algebra · PHD PROGRAM – U. DE CHILE
2025-II	Measure theory · MATHEMATICAL ENGINEERING – USACH
2025-I	Abstract algebra · MATHEMATICAL ENGINEERING – USACH
2025-I	Group theory and cellular automata · PHD PROGRAM – USACH
2024-II	Functional analysis · MATHEMATICAL ENGINEERING – USACH
2024-II	Measures and functional analysis · PHD PROGRAM – USACH
2024-I	Functional analysis · MATHEMATICAL ENGINEERING – USACH
2023-II	Measure theory · MATHEMATICAL ENGINEERING – USACH
2023-I	Real analysis · MATHEMATICAL ENGINEERING – USACH
2023-I	Hyperbolic groups · PHD PROGRAM – USACH
2022-II	Measure theory · MATHEMATICAL ENGINEERING – USACH
2022-I	Abstract algebra · MATHEMATICAL ENGINEERING – USACH
2022-I	Dynamical systems · PHD PROGRAM – USACH
2021-II	Measure theory · MATHEMATICAL ENGINEERING – USACH
2021-II	Algorithmic group theory · PHD PROGRAM – USACH
2021-I	Real analysis · MATHEMATICAL ENGINEERING – USACH
2021-I	Ergodic theory for group actions · PHD PROGRAM – USACH
2020-II	Linear algebra · MATHEMATICAL ENGINEERING – USACH
2020-I	Real analysis · MATHEMATICAL ENGINEERING – USACH
2018-II	Differential calculus · ECONOMY PROGRAM – UBC
2018-II	Calculus III · SCIENCE PROGRAM – UBC
2018-I	Calculus II · SCIENCE PROGRAM – UBC
2017-I	Calculus II · SCIENCE PROGRAM – UBC

PHD STUDENTS

2025–	Victor Valdebenito Silva PHD PROGRAM · USACH Defense expected 2027
2022–2025	Paola Rivera · co-directed with Cristóbal Rivas PHD PROGRAM · USACH Thesis: Grupos de automorfismos y G -subshifts.
2022–2024	Nicanor Carrasco Vargas · co-directed with Cristóbal Rojas PHD PROGRAM · PUC Thesis: Subshifts on groups and computable analysis

MASTER STUDENTS

2026–	Melanie Cubillos MASTER PROGRAM – USACH · Defense expected 2026
2024–2025	Paul Toussaint Browne MASTER PROGRAM – USACH · Thesis: Computability Notions for Quasi-isometries
2024–2024	Leo Poirier RESEARCH VISIT WITHIN M2 PROGRAM- ENS DE LYON ·
2022–2024	Esteban Cifuentes Alvarado MASTER PROGRAM · USACH Thesis: Problema 108 de Rufus Bowen

UNDERGRAD STUDENTS

2026–	Exequiel De la Cuadra MATHEMATICAL ENGINEERING PROGRAM – USACH · Defense expected 2026
2026–	Diego Cabrera MATHEMATICAL ENGINEERING PROGRAM – USACH · Defense expected 2026

NATURAL LANGUAGES

Spanish	mother tongue
English	● ● ● ● ●
French	● ● ● ● ●
Italian	● ● ● ● ●

COMPUTER LANGUAGES

LaTeX	● ● ● ● ●
Sage	● ● ● ● ●
Rust	● ● ● ● ●
Python	● ● ● ● ●
C++	● ● ● ● ●

